

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME:SARAH SHARMEEN R
COURSE CODE: Computer Networks-CSA0708

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks.
(BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.
Analyze the missing layers and explain how their functions are essential for reliable communication.

SOLUTION:

The OSI model has 7 layers:

1. Application
2. Presentation
3. Session ← Missing
4. Transport
5. Network
6. Data Link ← Missing
7. Physical

Completed Concept Map:

Application → Presentation → Session → Transport → Network → Data Link → Physical

Functions of the Missing Layers:

1. Session Layer

The Session Layer establishes, manages, and terminates communication sessions between applications.

Functions:

- Establishes and terminates sessions.
- Maintains communication between two devices.
- Provides synchronization using checkpoints.
- Helps recover communication if a connection is interrupted.

Importance: It ensures that communication between applications is properly maintained and can be resumed after interruptions.

2. Data Link Layer

The Data Link Layer provides reliable communication between two directly connected devices.

Functions:

- Converts packets into frames.
- Uses MAC addresses for physical addressing.
- Performs error detection.
- Controls access to the transmission medium.
- Helps ensure reliable frame delivery.

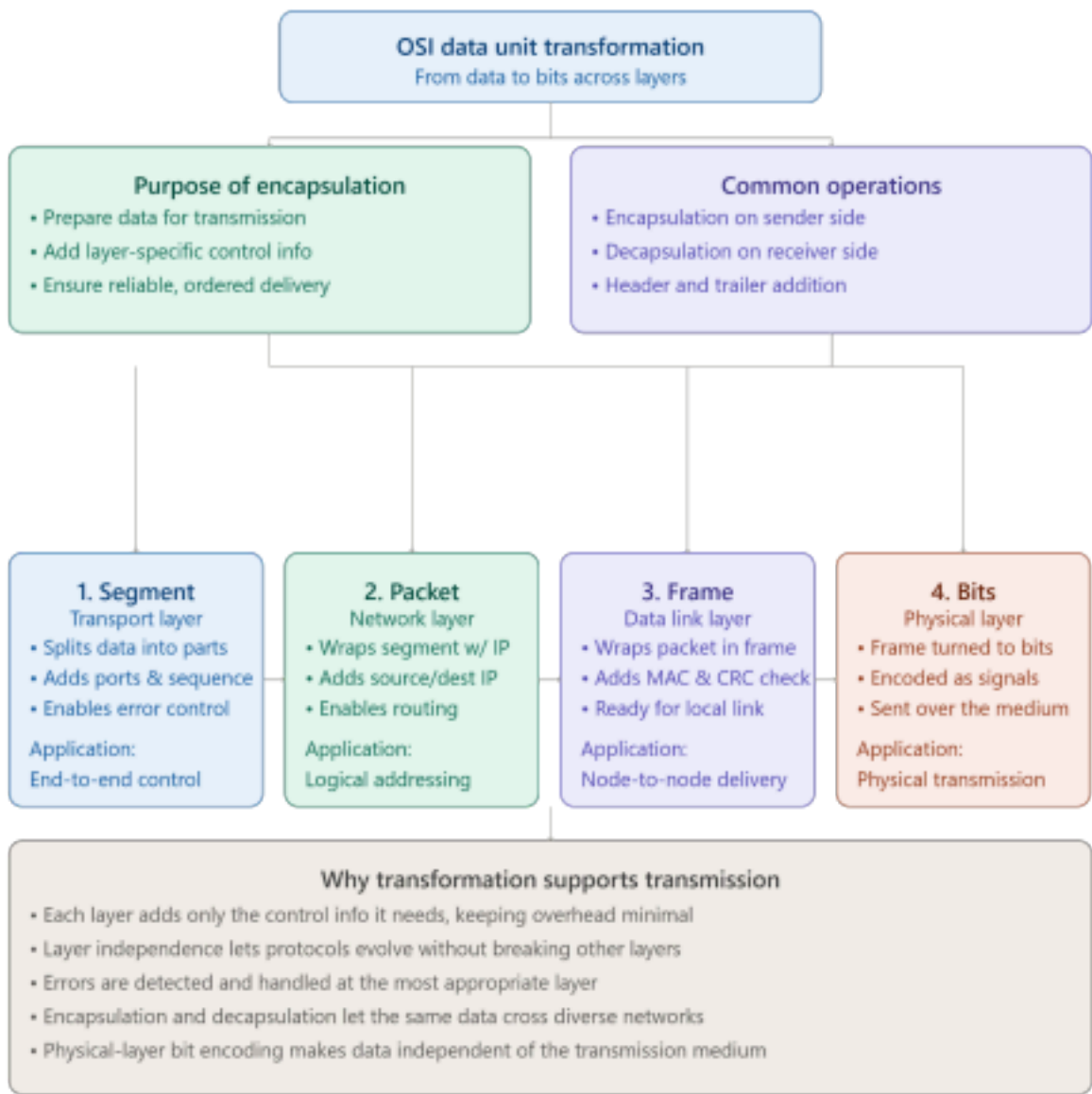
Importance: It ensures that data is transferred correctly between neighboring devices and detects errors that occur during transmission.

Conclusion:

The missing layers are Session and Data Link. The Session Layer manages and maintains communication sessions, while the Data Link Layer ensures reliable frame delivery and error detection. Together with the other OSI layers, they enable organized, reliable, and efficient end-to-end communication across a network.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

SOLUTION:



Data Unit Transformation

Data → Segment → Packet → Frame → Bits

- Transport Layer → Segment: Breaks data into smaller segments and adds transport-layer information such as port numbers.
- Network Layer → Packet: Adds IP addresses and creates packets for routing across networks.
- Data Link Layer → Frame: Encapsulates packets into frames and adds MAC addresses and error-detection information.
- Physical Layer → Bits: Converts frames into a stream of 0s and 1s and

transmits them as electrical, optical, or wireless signals.

Analysis: How Transformation Supports Transmission

As data moves down the OSI layers, each layer adds its own control information through a process called encapsulation. This allows the data to be properly addressed, routed, error-checked, and transmitted.

3. Given the concept map:

DataLink → MACAddress → LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

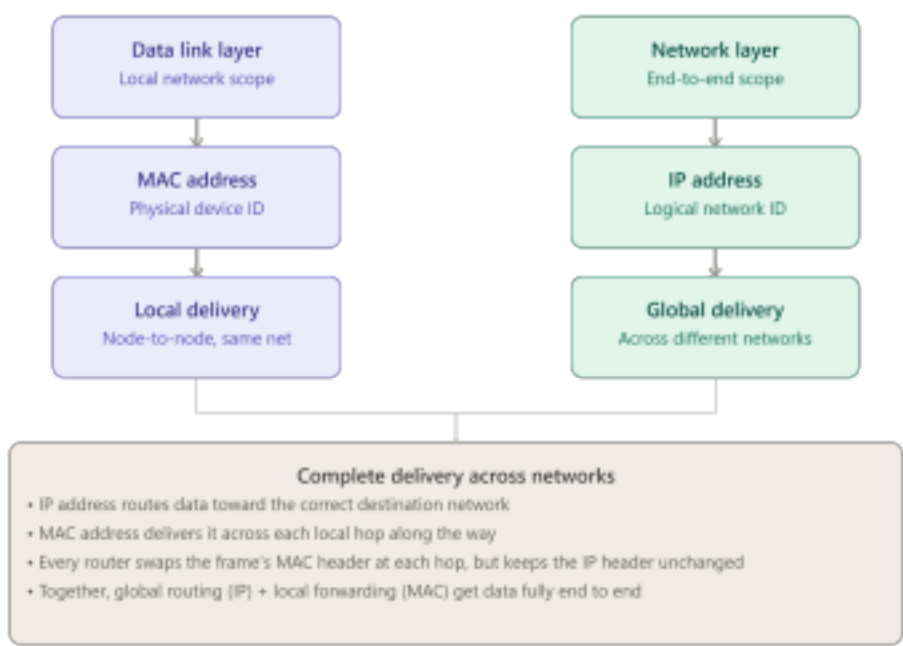
SOLUTION:

The Data Link Layer uses MAC (Media Access Control) addresses to deliver data between devices on the same local network (LAN). A MAC address uniquely identifies a network interface, allowing frames to be delivered to the correct device within the local network.

The Network Layer uses IP addresses to deliver data across different networks. The IP address identifies the source and destination devices and helps routers determine the best path for forwarding packets from one network to another.

How They Work Together:

1. The IP address identifies the final destination of the data across different networks.
2. Routers examine the destination IP address and forward the packet toward the correct network.
3. At each network, the Data Link Layer uses the MAC address to deliver the frame to the next device (such as a router or the final destination).
4. This process continues until the packet reaches the destination network.
5. Finally, the MAC address is used to deliver the frame to the destination device on the local network.



4. A concept map shows:

ErrorControl

→DataLinkLayer→ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

SOLUTION:

The Data Link Layer and Transport Layer work together to improve the reliability of data transmission, but they handle errors at different levels.

1. Data Link Layer – Error Detection

The Data Link Layer checks for errors that occur while data is being transmitted between directly connected devices. It uses techniques such as CRC (Cyclic Redundancy Check) to detect corrupted frames.

- Detects errors in frames.
- Discards corrupted frames.
- May request retransmission of damaged frames.

2. Transport Layer – Error Correction

The Transport Layer, especially TCP, provides reliable end-to-end communication between the source and destination.

- Detects missing or lost segments.
- Uses acknowledgements (ACKs) to confirm successful delivery.
- Retransmits lost or corrupted data.
- Ensures data arrives in the correct order.

How They Work Together:

Sender

↓

Data Link Layer: Detects errors in each frame

↓

Network: Routes packets across networks

↓

Transport Layer: Ensures complete and correct end-to-end delivery

↓

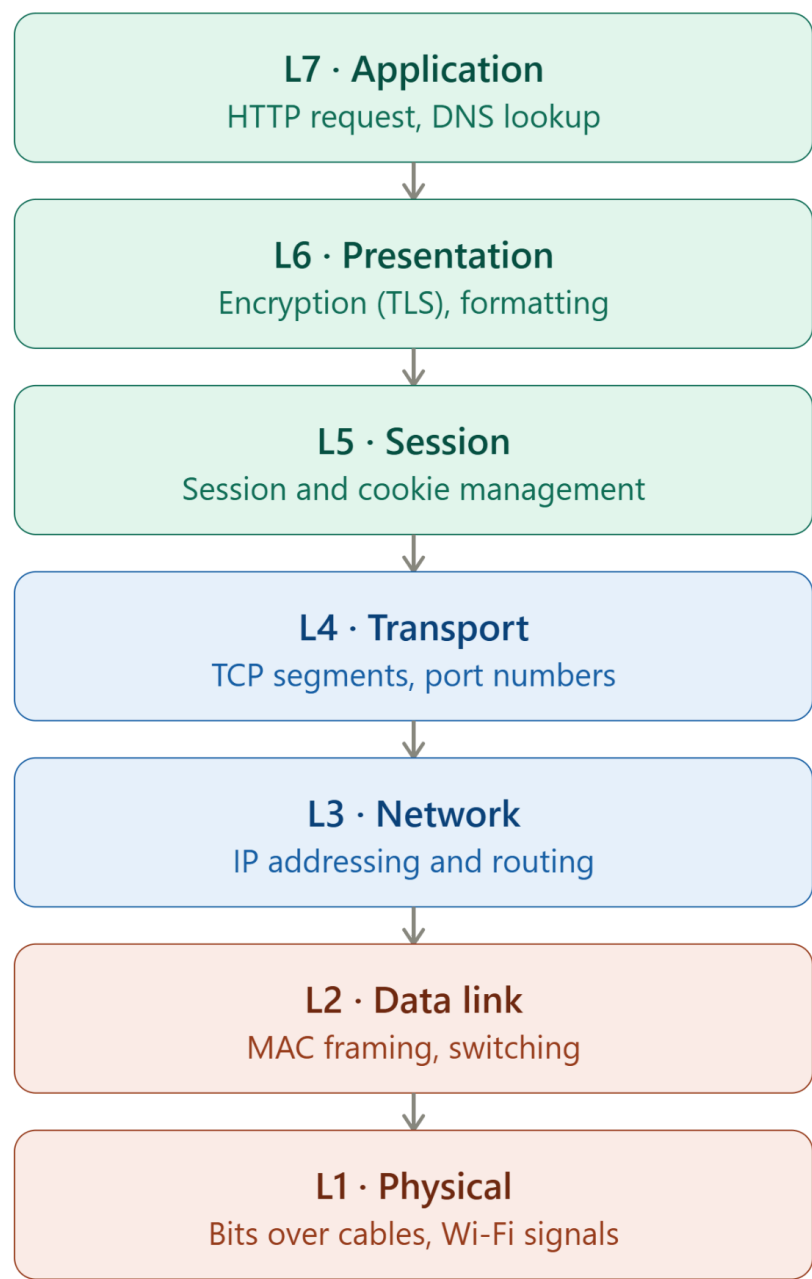
Receiver

Conclusion:

The Data Link Layer provides local error detection, while the Transport Layer provides end-to-end reliability through error recovery and retransmission. Their interaction ensures that errors are detected early at the link level and that missing or damaged data is recovered at the end-to-end level. This results in more reliable and accurate data transmission across networks.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

SOLUTION:



Teal = service layers (contained failure)
Blue = delivery layers
Coral = physical/link (total failure if down)

6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗

Analyze the problem and explain how the concept map helps in troubleshooting.

SOLUTION:

Analysproblem is fixed, the Transport and Application Layers can function normally again.is of the Problem:

The concept map shows that the Physical Layer and Data Link Layer are functioning correctly, but the Network Layer has failed.

- Physical Layer ✓: The physical connection, such as cable, Wi-Fi, or signal, is working.
- Data Link Layer ✓: Local communication and frame delivery are working correctly.
- Network Layer ✗: There is likely a problem with IP addressing, routing, or router configuration. Packets cannot be routed to their destination.
- Transport Layer ✗: Since packets cannot reach the destination, reliable end-to-end communication using protocols such as TCP cannot be established.
- Application Layer ✗: Applications such as web browsers or email cannot communicate with the required server.

How the Concept Map Helps in Troubleshooting:

The concept map helps identify the problem by showing the first failed layer. Since the Physical and Data Link layers are working, there is no need to immediately troubleshoot cables or local connections. The technician can focus on the Network Layer by checking:

1. IP address configuration.
2. Subnet mask and default gateway.
3. Router connectivity and routing tables.
4. Packet routing and IP configuration.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Marks Evaluation:

Total No. Question	Concept Identification (25%)	Linkages (25%)	Organization (20%)	Integration of Literature (20%)	Clarity (10%)	Total Marks (Each – 50)
6						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____